

HRISHIV KHATIWALA

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EDUCATION

Green Hope High School | Cary, NC | Expected June 2028

GPA: 4.696/5.00 Weighted · 3.96/4.00 Unweighted | SAT: 1430 | Class Rank: 7 of 652

AP Coursework: AP World History · AP Precalculus · AP Comparative Government ·

AP Physics 1 · AP Statistics

CompTIA ITF+ Certified (2024) | AoPS Advanced Mathematics (2022-2024)

Wake Technical Community College | Cary, NC | Concurrent Enrollment

GPA: 4.00 | Pursuing Associate of Science (Expected June 2028)

RESEARCH EXPERIENCE

Duke University | Durham, NC

Research Assistant | May 2026 - Present

- Processed cross-sectional LOVAMAP software data in MATLAB to quantify pore size, perimeter, and equivalent circular dimensions across 3D scaffold domains
- Refactored analytical codebase into modular, reusable functions to compute 2D pore shape metrics and characteristic length efficiently across large datasets

NC State University | Raleigh, NC

Research Intern | May 2026 - August 2026

- Conducted research with Professor Paul Liu applying machine learning and large language models (LLMs) to build Python predictive models for hurricane tracking
- Contributed to data pipeline development and model validation for meteorological forecasting

NC School of Science and Mathematics (NCSSM) | Durham & Morganton, NC

Summer Accelerator Researcher | Summer 2024 · Summer 2025

- Selected for two competitive residential research programs on separate campuses
- Designed a passive cooling system inspired by capillary action in tree bark structures (Biomimicry, Durham campus)
- Mastered advanced biotechnology lab protocols using research-grade equipment unavailable at home institution (Morganton campus)

PUBLICATIONS

Khatiwala, H. (2026). How Engineering-Driven Research on Health Literacy Has Evolved.

Zenodo. <https://doi.org/10.5281/zenodo.21779056>

WORK EXPERIENCE

KSB Investment | Investment Intern | June 2026 - September 2026

- Analyzed real estate investment opportunities, coliving scenarios, and macro market indicators
- Presented findings at business conferences representing KSB operations

Robotics for All | STEM Curriculum Intern | 2023 - Present (Remote)

- Developed K-8 robotics curriculum reaching 5,000+ students across the United States
- Prioritized accessibility and engagement for low-income and underrepresented youth

Web Development (Freelance) | 2023 - 2024

- Built websites for small businesses; applied self-taught SEO and analytics skills

- Generated \$1,000+ in client revenue

LEADERSHIP & ACTIVITIES

American Rocketry Challenge - 4-H Certified Team

Co-Founder & Co-President | 2024 - Present

- Placed 2nd of 1,001 teams nationally; earned Rookie Team of the Year and \$2,000 prize
- Designed custom 3D-printed fins optimized for aerodynamic stability and structural performance
- Built AI altitude prediction models from flight data and weather inputs; reduced error to under 10 feet
- Presented at National 4-H Congress and STEAM Camps; mentored peers through GHHS Aerospace Club

VEX Robotics - Gridwalkers (gridwalkers.net)

Co-Founder, Co-President & Mechanical Lead | 2025 - Present

- Co-founded team from scratch; secured \$1,000+ in competitive grants for equipment and fees
- Engineered robot intake and drivetrain for VEX V5 platform
- Achieved 12-0 record at Wolverine Regional; qualified for NC State Championship (top 36 teams statewide)
- Invited to VEX U.S. Open - top 150 teams worldwide; presented robot to industry executives

Science Olympiad - Green Hope High School

Varsity Member, Build Specialist & Helicopter Coach | 2022 - Present

- Set 2025 NC State Record in Helicopter: 2 min 26 sec with custom rubber-powered balsa-mylar aircraft
- 7th Place Helicopter at NC State Championship (2025); multiple 1st and 2nd place finishes across events (2022-2025)
- Coach middle school students in Helicopter and Flight events

NC DECA - Green Hope Chapter

Competitive Member | 2024 - Present

- Top 5% statewide at NC DECA State Competition; qualified for ICDC (International Career Development Conference)
- Authored 20-page business case study on PingPod (recognized as a top paper at Green Hope HS)
- Built a custom interactive web application to present the marketing strategy (pingpod.blog)
- Developed a 2-year marketing and growth strategy for a \$50M autonomous sports company

Read to Lead - Books for Africa

Co-Founder & Leader | 2022 - 2024

- Mobilized 1,400+ people to collect 10,000+ books for communities in rural Kenya and Tanzania
- Organized \$85,000+ in educational materials shipped across two continents
- Transformed a one-time drive into a recurring annual school tradition

Science of the Soul Study

Service Leader & Team Coordinator | 2013 - Present

- 350+ volunteer hours; facility hosts 3,000+ annual visitors
- Lead multilingual teams in infrastructure projects, logistics, and large-scale community events

AWARDS & HONORS

NC State Record - Helicopter Event (2 min 26 sec)

2nd Place of 1,001 Teams Nationally

Rookie Team of the Year

ICDC Qualifier - NC DECA State Top 5%

Science Olympiad, 2025

American Rocketry Challenge, 2024

American Rocketry Challenge, 2024

DECA, 2025

VEX U.S. Open Invitation - Top 150 Teams Worldwide	VEX Robotics, 2025
NCSSM Summer Accelerator Selectee (×2)	NCSSM, 2024 & 2025
PingPod Case Study - Top Paper, Green Hope High School	DECA, 2025

SKILLS & CERTIFICATIONS

Engineering & Design Aerodynamics · 3D printing & fin fabrication · biomimicry design · materials analysis · 3D modeling · VEX V5 robotics systems

Programming & Technical Python · MATLAB · HTML/CSS · JavaScript · AI/ML modeling · LOVAMAP software · flight simulation · data pipeline development

Research & Analysis Scientific writing · statistical analysis · lab protocols · quantitative data processing

Business & Communication Market strategy · case study writing · business presentations · web application development · SEO & analytics

Languages English (fluent) | Gujarati (native)

Certifications CompTIA ITF+ (2024) | 4-H Certified Team Leader